SILAS M. FALDE

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EDUCATION

University of Michigan School of Engineering **Ann Arbor, MI**
Junior B.E.S. in Data Science, GPA: 3.09/4.00 *August 2023 – December 2026*

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

Western High School **Parma, MI**
GPA: 3.963 *August 2019 – May 2023*

- Men's Varsity Soccer – Captain 1 year, Member 2 years
- Select Choir – Member 3 years

WORK EXPERIENCE

Ally Financial, Inc. **Detroit, MI**
Intern, Artificial Intelligence & Advanced Analytics *May 2026 – August 2026*

- Experimenting with large language models for task automation and improvement to enhance company efficiency and safety.
- Building NLP pipelines using Python to assess the performance of PII masking and redaction models, ensuring sensitive information is protected in company documents.
- Enhancing the functionality of a data dictionary migration tool by implementing batch processing, more intelligent metadata selection, and streamlined interface, resulting in improved usability and efficiency for data management tasks.

Ally Financial, Inc. **Detroit, MI**
Intern, Data Management *May 2025 – August 2025*

- Created data dictionaries and data lineage documentation leveraging Python, enhancing data governance and compliance.
- Added data quality checks to existing data pipelines using Python and SQL, improving data integrity and reliability.
- Developed a data availability dashboard in Power BI to monitor data refresh rates and availability, ensuring timely access to critical data.

RESEARCH

University of Michigan Department of Epidemiology **Ann Arbor, MI**
Research Assistant – Computational Lead *September 2025 – May 2026*

- **Lead** the technical development of a **Python** pipeline using a large language model to classify narratives of collective violence.
- Performed data cleaning, exploratory data analysis including **Natural Language Processing**, and descriptive statistical analysis on a large, sensitive dataset.
- Evaluated model performance using **performance metrics** to assess the accuracy of the classification model, identify areas for improvement, and suggest potential solutions to enhance its performance.

University of Michigan Department of Epidemiology **Ann Arbor, MI**
Research Assistant *September 2024 – May 2025*

- Awarded 3rd place (top ∼1%) in the CDC Youth Mental Health: Novel Variables Data Challenge.
- Developed **Python** pipelines using a large language model on a high performance computing cluster to extract trends from the narratives of victims of suicide.
- Cleaned, preprocessed, and **feature engineered** a confidential dataset of over 400,000 narratives.
- Documented findings with reports and visualizations to communicate findings to both the research team and the academic community through publication.

PROJECTS

FPS PvP Dynamics **Ann Arbor, MI**
Project Developer *March 2026 – April 2026*

- Built a configurable **agent-based simulation** in Python to model player behavior, combat, and objectives on a grid with stochastic state updates, rule-based decisions, and per-tick match tracking.
- Developed data collection and analysis tooling with **trace capture**, **CSV export**, and **visualization** for debugging and validation.
- Ran **parameter sweeps**, replication studies, **Latin hypercube sampling**, and **Sobol sensitivity analysis** to evaluate behavior and key parameters.

Photo Processing Helper **Ann Arbor, MI**
Project Developer *Jan 2026 - Present*

- Built a Python **image-processing** toolkit to automate various photographic workflows including photo framing, collage generation, panorama stitching, and RAW-to-JPG matching across large image folders.
- Implemented **computer vision** pipelines with OpenCV, NumPy, Pillow, and rawpy to resize, crop, align, and blend images while preserving exact output dimensions and high-bit-depth detail; added validation and testing around file discovery, output integrity, and stitching behavior to keep the pipelines reliable for real-world photo datasets.
- Designed a **modular** codebase with shared library logic, CLI tools, and a notebook workflow to support repeatable batch processing and interactive experimentation.

SKILLS

Programming	C++, Python, R, SQL, Java
Python	PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, NLTK, spaCy
Software	Git, VS Code, Microsoft Office Suite, Google Suite, Snowflake, Collibra, Alation, Manta